

Module 09 Aggregating Data

IBM InfoSphere Information Server for Data Integration



Module Objectives

After completing this topic, you should be able to:

- Aggregate data using Aggregate Stage

Properties of Aggregator Stage

- Aggregator Stage gives you the option of grouping and summary operations
- Records can be grouped by one or more columns
- Supports only one input link
- In some cases, the Hash Partitioning method must be used to avoid erroneous calculation
- Multiple aggregations can be computed using single Aggregation stage

Aggregation Example

Input File

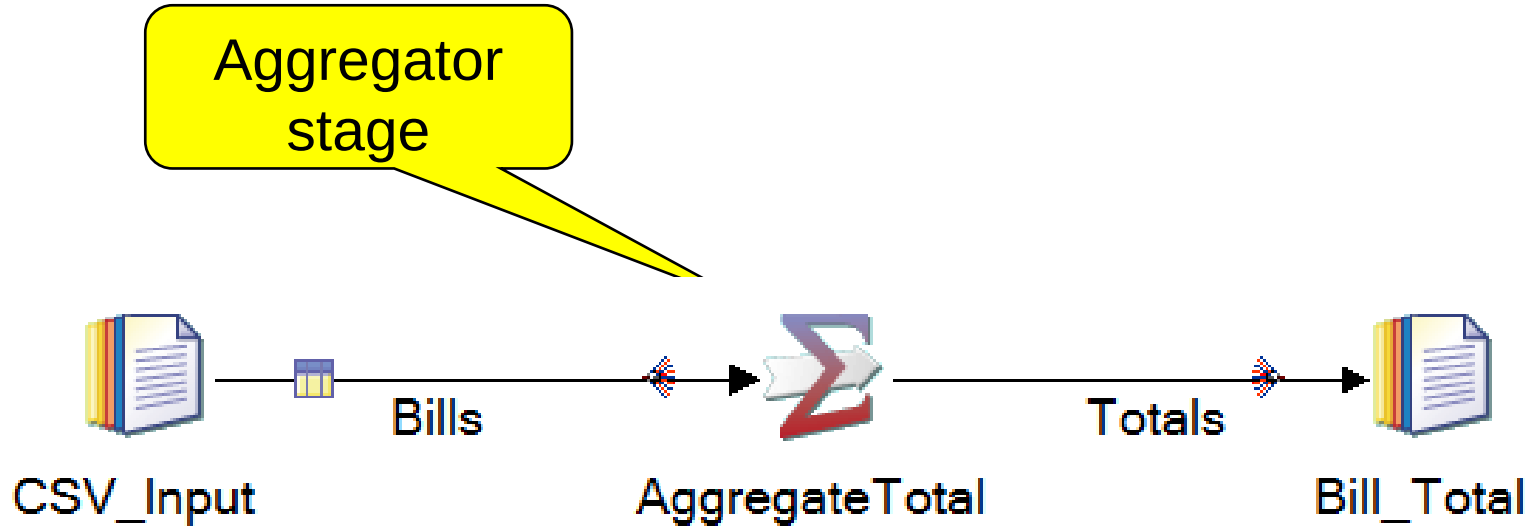
Country	State	County	Population in Mil.
USA	CA	Riverside	2.2
USA	CA	Alameda	1
USA	TX	Dallas	2.2
USA	TX	El Paso	0.7
USA	MA	Middlesex	1.5
USA	...		...

Grouping Key

Output File

Country	State	Population in Mil.
USA	CA	38
USA	MA	7
USA	TX	26
USA	...	...

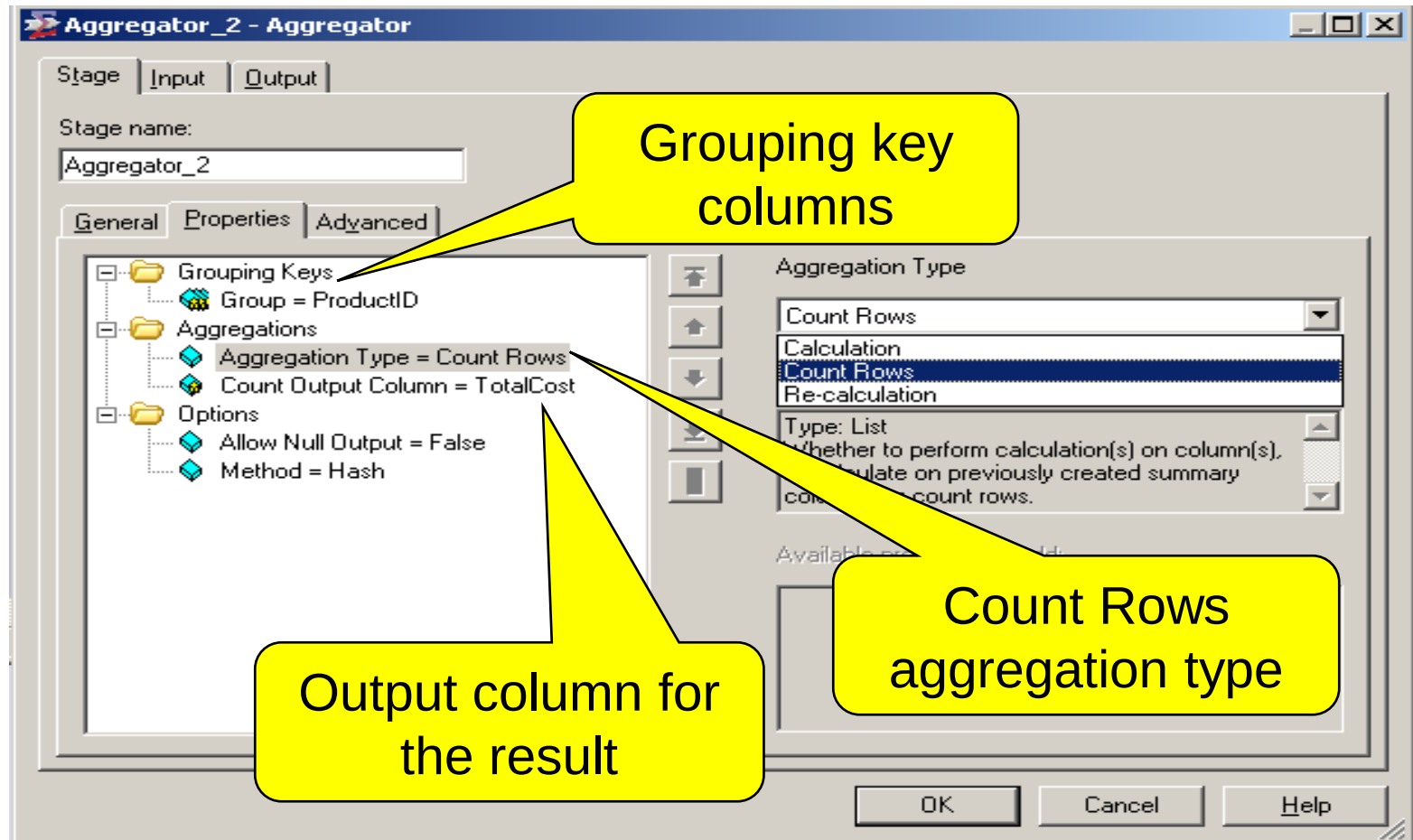
Job with Aggregator Stage



Aggregation Types

- Count rows
 - Count rows in each group
 - Put result in a specified output column
- Calculation
 - Select column
 - Put result of calculation in a specified output column
 - Calculations include:
 - Sum, Count, Min, max, Mean
 - Missing value (e.g. NULL) count
 - Non-missing value count
 - Percent coefficient of variation
- Recalculation
 - Aggregates previously summarized data
 - Treats the output column of a aggregation stage as input
 - Can be used for reiterated aggregations (e.g. first, take the sum of state population, then sum up the national population)

Aggregation Type: Count Rows



Aggregation Type: Calculation

Grouping key columns

Calculation aggregation type

Calculations with output columns

Available calculations

Aggregator_Count_Rows_Items - Aggregator

Stage | Input | Output

Stage name: Aggregator_Count_Rows_Items

General | Properties | Advanced

Grouping Keys

- Group = Item
- Case Sensitive = False
- Group = Warehouse

Aggregations

- Aggregation Type = Calculation
- Column for Calculation = Item
- Maximum Value Output Column = ItemMax
- Minimum Value Output Column = ItemMin

Options

- Allow Null Output = False
- Method = Sort

Column for Calculation: Item

Information:

Available properties to add:

- Corrected Sum Of Squares Output Column
- Mean Value Output Column
- Missing Value
- Missing Values Count Output Column
- Non-Missing Values Count Output Column
- Percent Coefficient Of Variation
- Preserve Type
- Range Output Column
- Standard Deviation Output
- Standard Error Output Column

OK Cancel Help

Aggregation Type: Calculation Example

Input data

Country	State	County	Population in Mil.
USA	CA	Riverside	2.2
USA	CA	Alameda	1
USA	TX	Dallas	2.2
USA	TX	El Paso	0.7
USA	MA	Middlesex	1.5
USA	...		...

Calculation Output and Re-calculation Input File

Country	State	Population in Mil.
USA	CA	38
USA	MA	7
USA	TX	26
USA	...	...

The screenshot displays the configuration interface for an aggregation operation. The left pane shows a tree structure with the following settings:

- Grouping Keys:**
 - Group = Country
 - Group = State
- Aggregations:**
 - Aggregation Type = Calculation
 - Column for Calculation = Population_State
 - Sum Output Column = Population
 - Summary Output Column = Summary** (highlighted with a blue selection bar)
- Options:**
 - Allow Null Output = False
 - Method = Hash

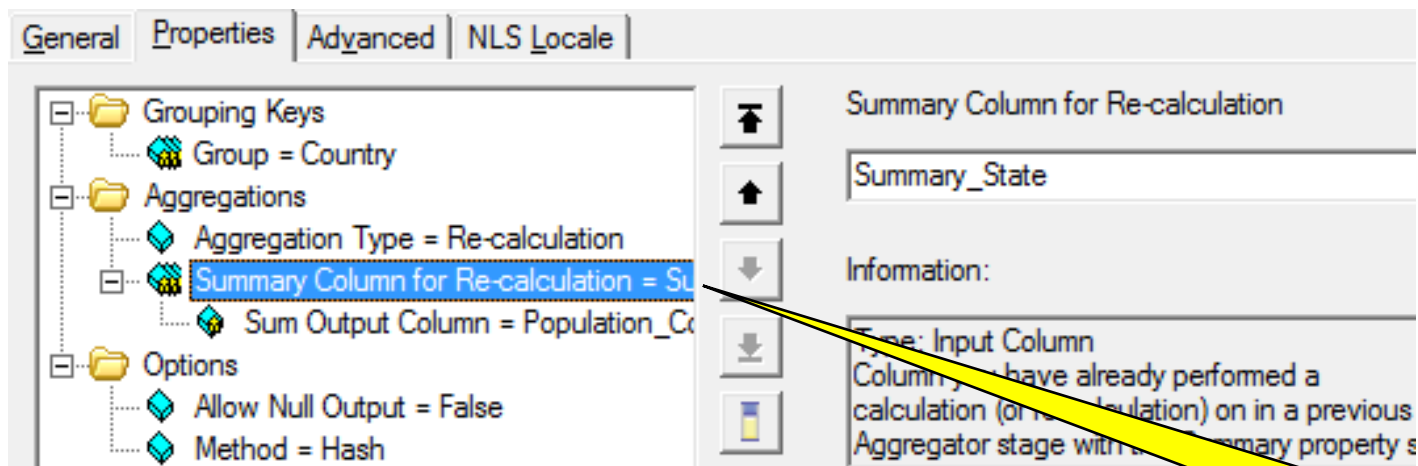
The right pane shows the 'Summary Output Column' field set to 'Summary_State'. Below it, the 'Information' section provides details about the 'Summary' property:

Type: Output Column
Name of subrecord column to which to write the results of the reduce or rereduce operation.

A yellow callout box labeled 'Summary property' points to the 'Summary Output Column = Summary' setting in the left pane.

Aggregation Type: Re-calculation Example Cont'd

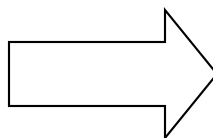
In short, re-calculation applies aggregate functions to a column that has already been summarized.



Summary Column
from previous stage

Re-calculation Output File

Country	State	Population in Mil.
USA	CA	38
USA	MA	7
USA	TX	26
USA	...	...

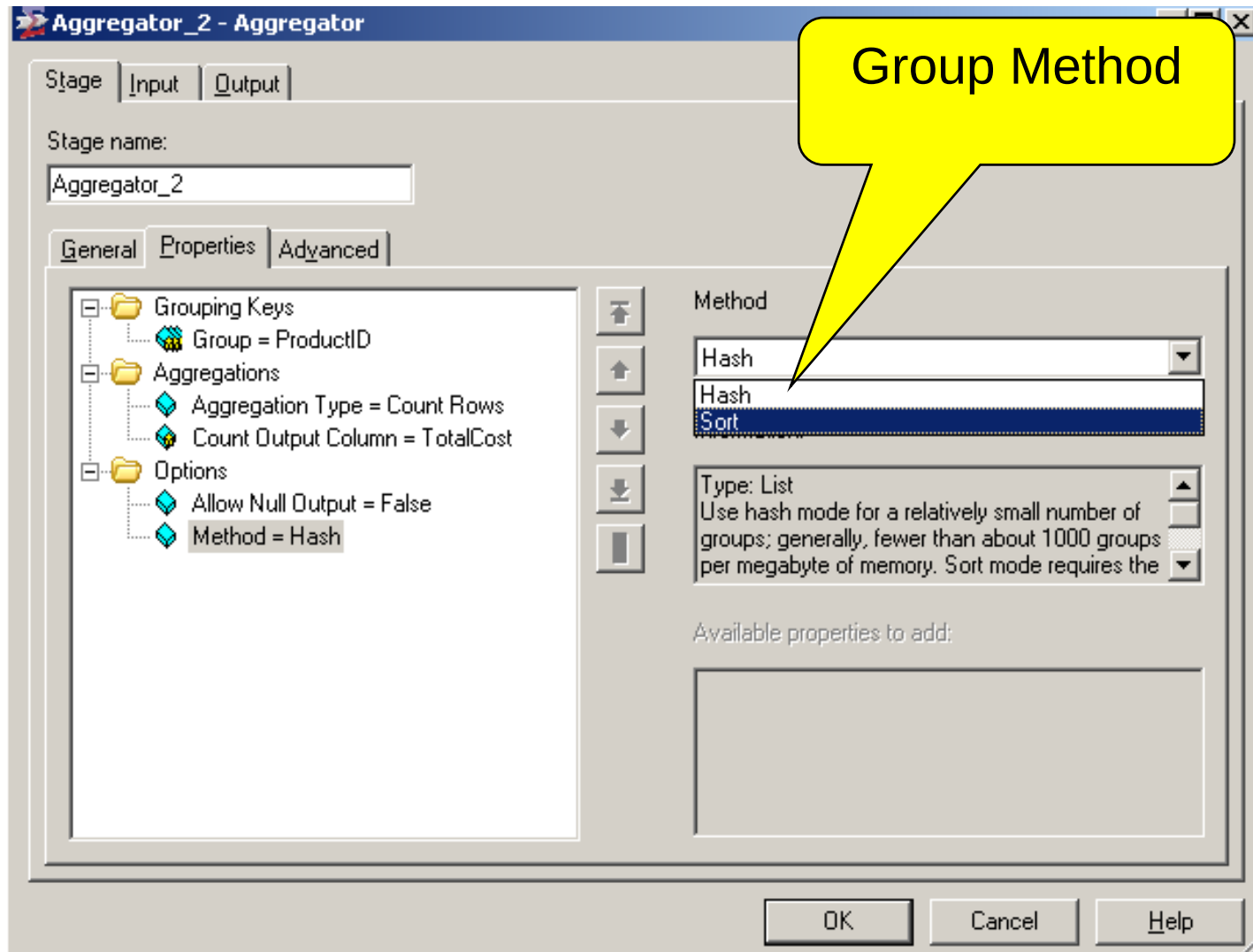


Sum Country	Population in Mil.
USA	313

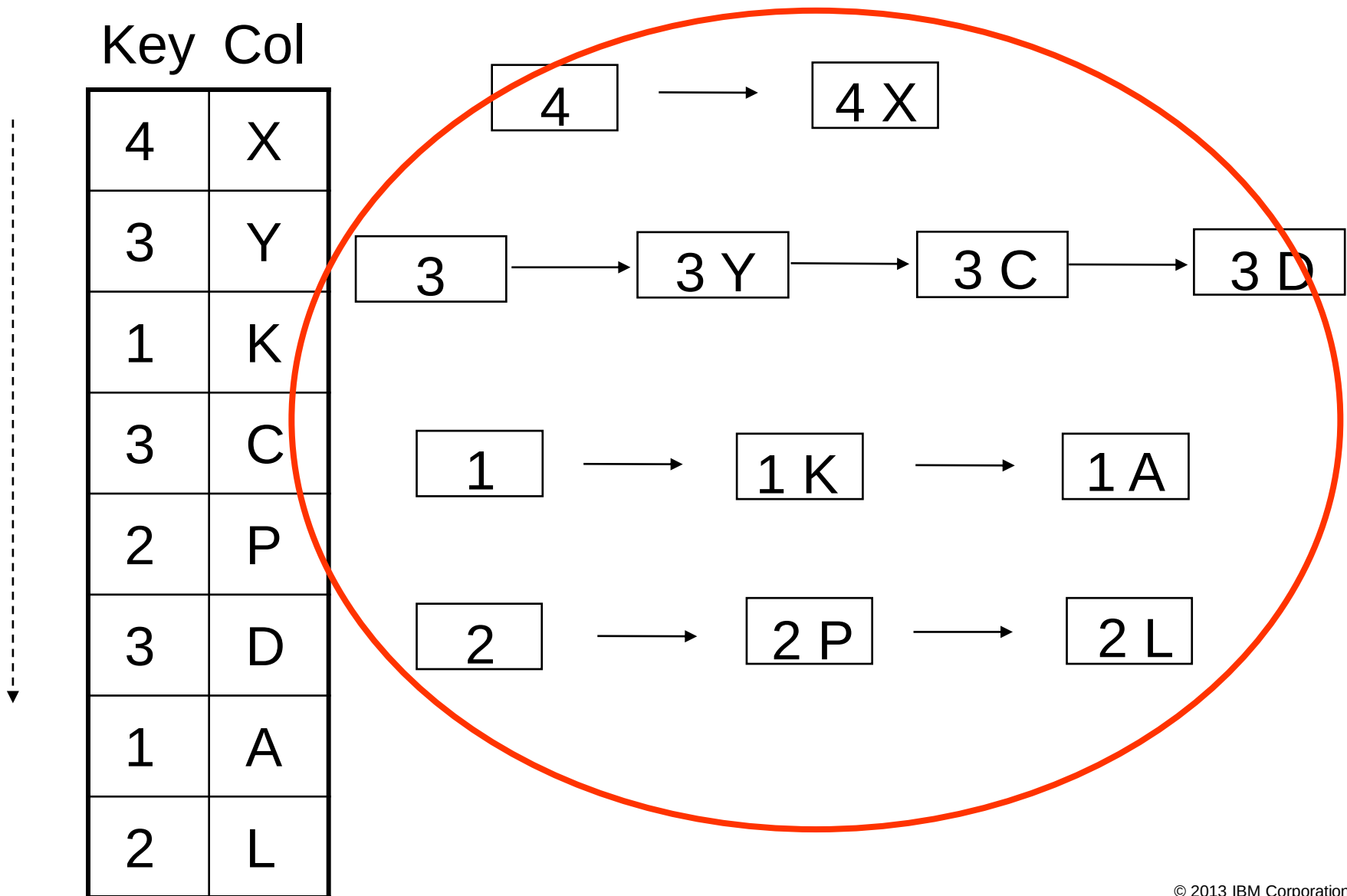
Grouping Methods

- Choice of grouping method should consider:
 - Number of distinct key values in the input set of records that will be used for grouping
 - Available memory
- Two Group methods:
 - Hash (default)
 - Calculations are made for all groups and stored in memory
 - Hashes the table structure
 - Results are written out after all input have been processed
 - Input does not need to be sorted
 - Useful when the number of unique groups is small
 - Running total for each group's aggregations needs to fit into memory
 - Sort
 - Treats the output column of a aggregation stage as input
 - Can be used for reiterated aggregations (e.g. first, take the sum of state population, then sum up the national population)
 - Requires the input data to be sorted by grouping keys
 - Does not perform the sort!
 - Only a single aggregation group is kept in memory
 - When a new group is seen, the current group is written out
 - Can handle unlimited numbers of groups

Group Method

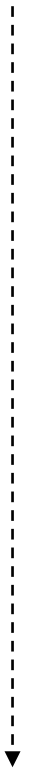


Grouping Method - Hash

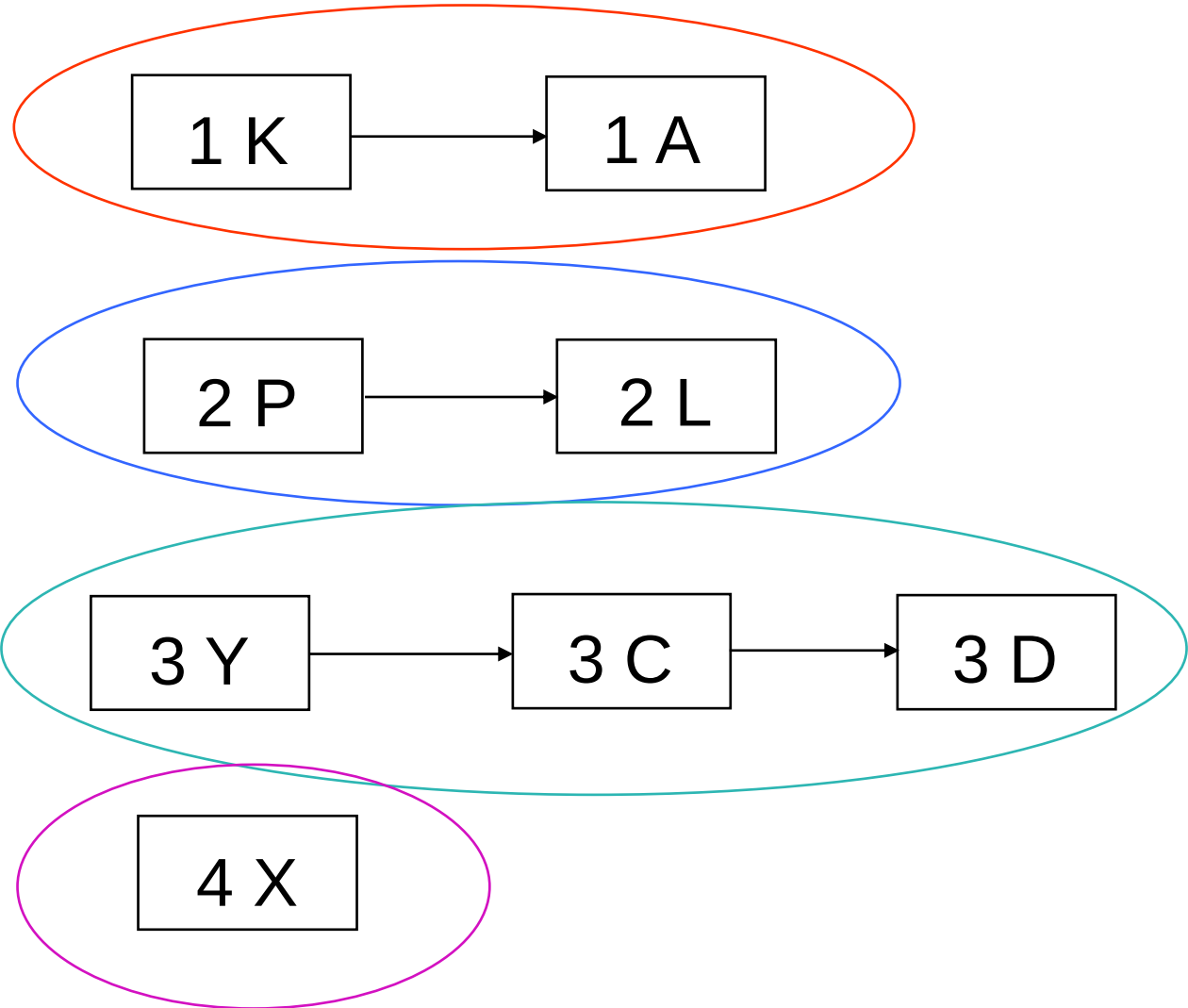


Grouping Method - Sort

Key Col

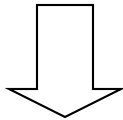


1	K
1	A
2	P
2	L
3	Y
3	C
3	D
4	X



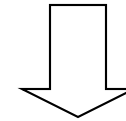
Grouping Method – Lessons Learned

1 Hash method holds all records in memory



Memory intensive

Sort method pre-sorts all records



Time intensive

2 Another implication involves the parallel processing. The pipeline parallelism are several processes busy at the same time. **The Hash method interrupts the pipeline flow of the records, since it holds everything into memory** and produces the output all at once. So using the hash has a performance implication as well.